VOL. I · NO. 1 MONDAYS & THURSDAYS

THE LAUNCH REPORT

Rocket launches, reentries and the business of spaceflight, worldwide

MONDAY, OCTOBER 12, 2026 ALL TIMES EASTERN DAYLIGHT TIME (EDT)

THE WEEK IN LAUNCH

The SDA batch finally flies, China adds 27th SatNet group, and three old stages come home

A SpaceX Falcon 9 carried 21 Space Development Agency (SDA) satellites to orbit from California at 3:39 a.m. EDT Saturday, five days after an abort at T-minus 58 seconds. Eleven hours earlier, at 3:27 p.m. EDT Friday, a Long March 12 lifted off from Hainan with the 27th group of China's SatNet low-orbit internet satellites. Those were the world's only two orbital launches since Thursday, according to Launch Library 2, bringing 2026 to 239 attempts.

All three rocket stages predicted to reenter on Saturday, including a Delta 2 stage launched in 1990, did so, according to CelesTrak. This week brings a Starlink launch Monday night, NASA's CRS-35 cargo flight to the International Space Station (ISS) on Tuesday and a U.S. Space Force mission on Thursday, with Japan's H3 and its Mars-moon probe a week out. Elon Musk said Starship's next flight is four to six weeks away and will try to catch the ship itself.

I · RECENT LAUNCHES & RETURNS

Two launches, two successes

Falcon 9 cleared a week of delays for the SDA; China's Long March 12 flew its fifth mission of the year.



A Falcon 9 lifts off from Space Launch Complex 4E at Vandenberg Space Force Base with NASA's SWOT satellite on Dec. 16, 2022. Saturday's SDA launch flew from the same pad.

PHOTO: NASA/KEEGAN BARBER · PUBLIC DOMAIN

Falcon 9: SDA Tranche 1 Transport Layer A

Falcon 9 lifted off from Space Launch Complex 4E at Vandenberg Space Force Base, California, at 3:39 a.m. EDT on Saturday, October 10 (07:39 UTC), with 21 Northrop Grumman-built satellites for the SDA's Tranche 1 Transport Layer, Spaceflight Now reported. Booster B1103, on its sixth flight, landed at Landing Zone 4 about 7.5 minutes after liftoff.

The first attempt, on October 5, stopped 58 seconds before liftoff after onboard computers flagged "some off-nominal data on Stage Two." SpaceX then moved the date a day at a time. "Additional time throughout the week allowed for double checks on vehicle systems and range deconfliction," SpaceX launch commentator Atticus Vadera said. Spaceflight Now said the range conflict is believed to have been a missile test from Vandenberg on Friday; the Federal Aviation Administration's (FAA) advisory had listed a Northrop Grumman MK21-2B launch from the base that morning. It was the fourth of six Tranche 1 Transport Layer launches.

Long March 12: SatNet LEO Group 27

China launched a Long March 12 from the commercial launch site in Hainan at 3:27 a.m. Beijing time on Saturday, 3:27 p.m. EDT Friday, and placed the 27th group of low-orbit internet satellites into the preset orbit, Xinhua reported. The flight from Commercial Launch Complex 2 at Wenchang was within the 3:20–3:49 p.m. EDT window that airspace notices had shown, and China Aerospace Science and Technology Corp. (CASC) did not release the number of satellites. Launch Library 2 records it as a success. SatNet, also known as Guowang, is China's state-run broadband megaconstellation.

Three Falcon 9s this week, then H3 to Mars

Every orbital launch with a date between now and October 26, plus missions targeting later this month. Times are EDT, with Coordinated Universal Time (UTC) below; NET means no earlier than.

DATE / TIME	VEHICLE	PROVIDER	MISSION / PAYLOAD	SITE	ORBIT	STATUS	SOURCES
Mon Oct 12, 8:11 p.m. 00:11 UTC Oct 13	Falcon 9	SpaceX	Starlink 15-25: 27 Starlink V2 Mini satellites. Booster B1088, 20th flight	SLC-4E, Vandenberg, Calif.	LEO, 70°	**TENTATIVE**	Launch Library 2; Spaceflight Now
Tue Oct 13, 6:33 a.m. 10:33 UTC	Falcon 9 / Cargo Dragon	SpaceX for NASA	CRS-35: more than 6,300 lb of cargo for the ISS. Booster B1096, 8th flight	SLC-40, Cape Canaveral, Fla.	LEO (ISS)	**CONFIRMED**	NASA; Launch Library 2
Thu Oct 15, 7:44 p.m. 23:44 UTC	Falcon 9	SpaceX for the U.S. Space Force	USSF-481. Booster B1081, 28th flight	SLC-4E, Vandenberg, Calif.	Polar LEO	**TENTATIVE**	Launch Library 2 (from maritime warnings)
Oct 15–20 target	GSLV Mk II	Indian Space Research Organisation (ISRO)	NVS-03 (IRNSS-1L) navigation satellite	Second Launch Pad, Sriharikota, India	GTO	**PRELIMINARY**	Times of India; Launch Library 2
Mon Oct 19, 3:41:03 p.m. 19:41:03 UTC	H3-24	Mitsubishi Heavy Industries (MHI) for JAXA	Martian Moons eXploration (MMX), Phobos sample-return probe	LP-2, Yoshinobu complex, Tanegashima, Japan	Mars transfer	**CONFIRMED**	JAXA launch plan; Launch Library 2
NET Oct 29	Vulcan VC6L	United Launch Alliance (ULA)	Amazon Leo LV-01 broadband satellites	SLC-41, Cape Canaveral, Fla.	LEO	**PRELIMINARY**	Launch Library 2
NET October	Ariane 64	Arianespace	Amazon Leo LE-04	ELA-4, Kourou, French Guiana	LEO	**PRELIMINARY**	Launch Library 2
NET October	Falcon 9	SpaceX	Bandwagon-5 rideshare	SLC-40, Cape Canaveral, Fla.	Mid-inclination LEO	**PRELIMINARY**	Launch Library 2
NET October	Long March 12A	CASC	Flight 2	Jiuquan, China	LEO	**PRELIMINARY**	Launch Library 2
NET October	SSLV	ISRO	AayulSAT-1A and others	First Launch Pad, Sriharikota, India	LEO	**PRELIMINARY**	Launch Library 2

CONFIRMED date and time announced by the launch provider, its customer or the FAA. **TENTATIVE** specific date in published schedules or airspace and maritime notices, not yet confirmed by the operator. **PRELIMINARY** target range, or no-earlier-than date or month only.

Abbreviations: SLC, Space Launch Complex; SLC-4E, Space Launch Complex 4 East; LP-2, Launch Pad 2; ELA-4, Ariane Launch Area 4; LEO, low Earth orbit; GTO, geostationary transfer orbit; GSLV, Geosynchronous Satellite Launch Vehicle; SSLV, Small Satellite Launch Vehicle; JAXA, Japan Aerospace Exploration Agency.

Date changes

SDA Tranche 1 Transport Layer A. Set for Oct 5 (abort at T-58 seconds), then Oct 6, 8, 9 and 10; flew Sat Oct 10 at 3:39 a.m., ten minutes after the 3:29 a.m. time listed Thursday. SpaceX cited checks of vehicle systems and range deconfliction (Spaceflight Now).

Starlink 15-25. First listed for Sep 30, then Oct 10 and Oct 11; now Monday evening. Spaceflight Now lists a window opening at 7:00 p.m.; Launch Library 2 lists 8:11 p.m. No reason given.

NVS-03. Unchanged window of October 15 to 20, with no firm date or time from ISRO (Times of India).

What's flying

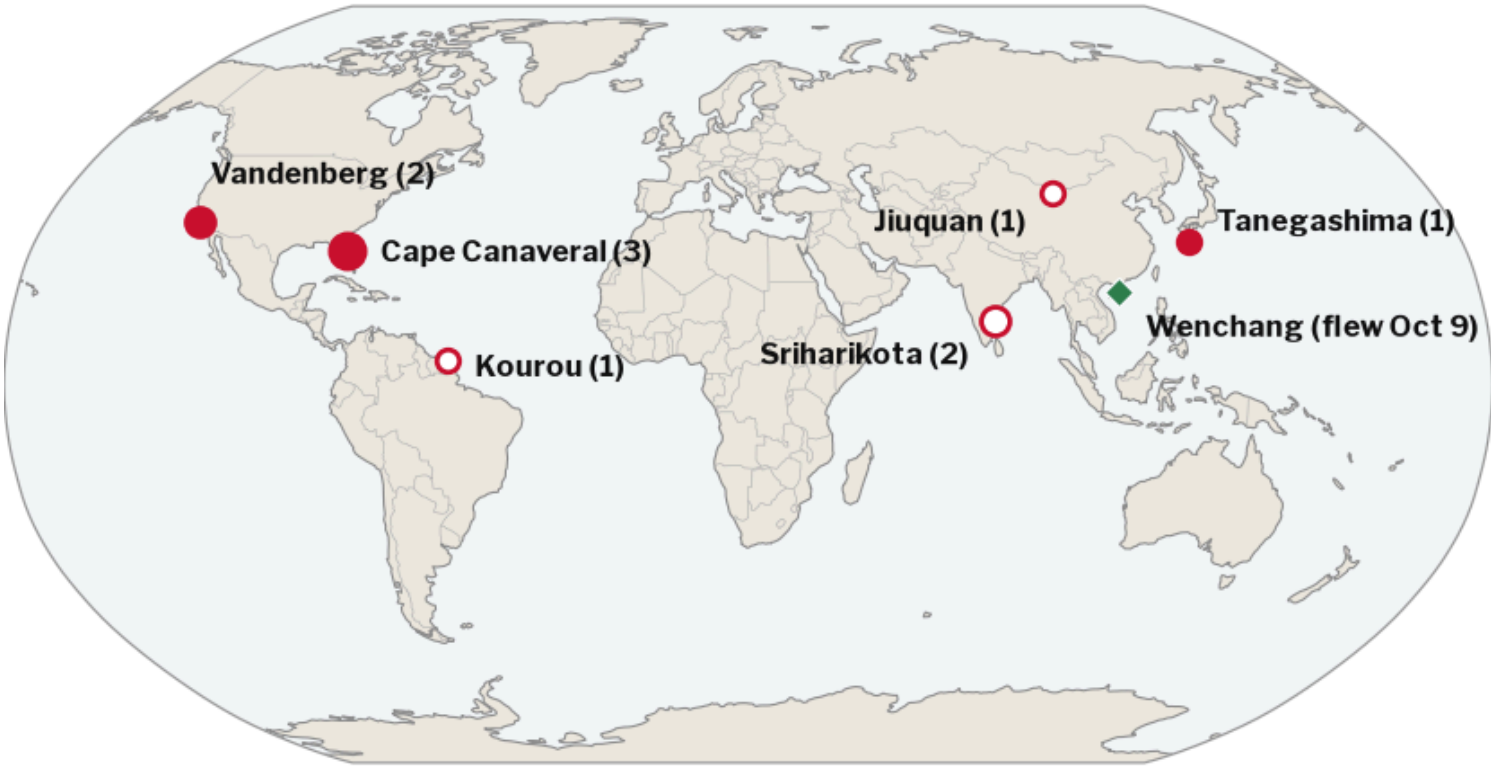
CRS-35. The 35th SpaceX cargo flight to the ISS uses Cargo Dragon C211 on its fourth flight and carries the final pair of ISS Roll-Out Solar Arrays. Docking at the space-facing port of the Harmony module is planned for Thursday, October 15 (Spaceflight Now).

H3 / MMX. Liftoff is 4:41:03 a.m. Japan Standard Time on Tuesday, October 20; the reserve period runs to November 7 (JAXA launch plan).

QUICK HIT

Starship. Musk said Flight 15 is "probably about four weeks or so" away and will attempt to catch the ship as well as the booster, TSN Media reported. NASASpaceflight's unofficial schedule shows NET November 8 from Starbase Pad 2.

Vulcan. ULA's return to flight with Amazon Leo satellites remains NET October 29 in Launch Library 2.



Launch sites with orbital launches scheduled through October 31; numbers count launches. Filled: dated launch (Confirmed or Tentative). Hollow: Preliminary. Green diamond: Wenchang, where the Long March 12 flew on October 9. Vandenberg also launched on October 10.
MAP: THE LAUNCH REPORT · PAD COORDINATES FROM LAUNCH LIBRARY 2 · NATURAL EARTH

III · TRAJECTORIES & HAZARD AREAS

Where H3's spent stages will fall

JAXA has published the sea areas where MMX's boosters, nose fairing and first stage are expected to come down.

H3 Flight 10 timeline

According to the launch plan, the rocket turns its pitch plane to an azimuth of 90 degrees, due east, shortly after liftoff and flies out over the Pacific. The upper stage fires twice and releases MMX into a Mars transfer orbit 71 minutes after liftoff. The times below assume launch on October 20 JST. Because the probe is headed to Mars, events after the first second-stage cutoff shift with the launch date.

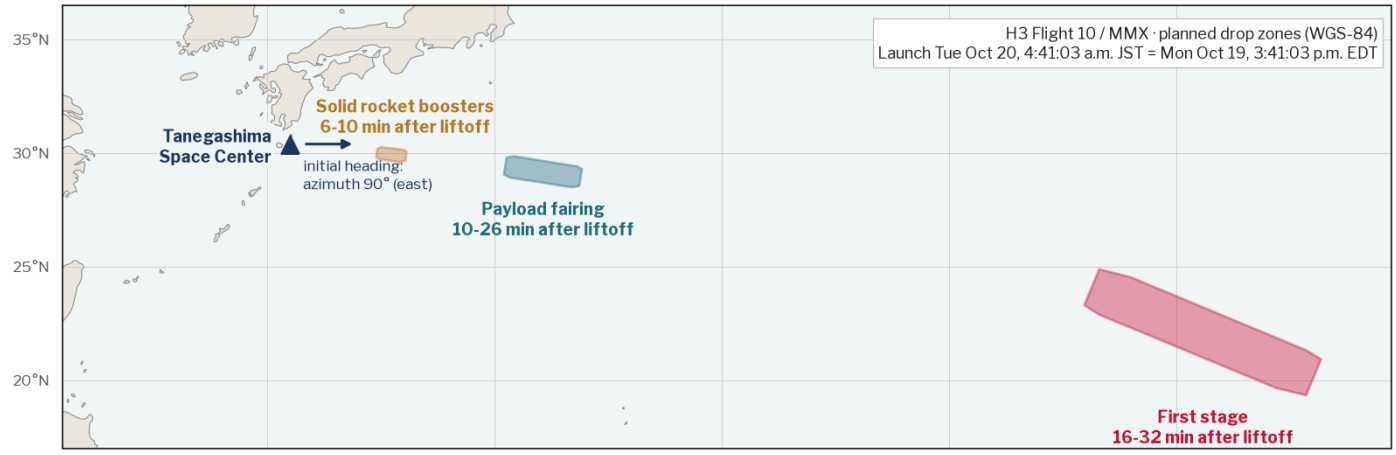
EVENT	TIME AFTER LIFTOFF	ALTITUDE
Solid rocket boosters separate (two pairs)	1:56 / 1:58	59–61 km
Payload fairing separates	3:17	131 km
Main engine cutoff (MECO)	4:52	208 km
First and second stages separate	4:59	214 km
Second-stage engine start 1 (SELI1)	5:12	224 km
Second-stage engine cutoff 1 (SECO1)	10:40	329 km
Second-stage engine start 2 (SELI2)	1:03:57	322 km
Second-stage engine cutoff 2 (SECO2)	1:10:21	497 km
MMX separates, 11.2 km/s	**1:11:12**	**592 km**



JAXA's HTV-X1 cargo spacecraft approaches the International Space Station on Oct. 29, 2025, four days after launching from Tanegashima on an H3-24, the same H3 version that will carry MMX.
PHOTO: NASA · PUBLIC DOMAIN

This week's U.S. launches

No hazard-area coordinates or ground tracks have been published for this week's Falcon 9 launches. From Vandenberg, Starlink 15-25 is headed for a 70-degree orbit and USSF-481 for a polar orbit, according to Launch Library 2.



Planned drop zones for H3 Flight 10, plotted from the corner coordinates in JAXA's launch plan (World Geodetic System 1984). The solid rocket booster zone measures 62 by 128 km, the fairing zone 107 by 330 km and the first-stage zone 226 by 1,105 km. Ships and aircraft are warned to stay out of these zones through Japan's coast guard and transport ministry. The arrow shows the eastward heading the plan gives for shortly after liftoff; it is not a ground track.

MAP: THE LAUNCH REPORT · DATA FROM THE JAXA H3/MMX LAUNCH PLAN (AUG. 20, 2026) · NATURAL EARTH

IV · REENTRIES & ISS TRAFFIC

All three predicted stages came down Saturday

A Delta 2 first stage launched in 1990 ended 36 years in orbit. At the station, CRS-35 arrives Thursday.

Confirmed reentries

CelesTrak's satellite catalog now lists all three rocket bodies that The Aerospace Corporation had predicted, with uncertainties of 14 to 16 hours, for Saturday, October 10, as decayed that day. Exact times and locations have not been published.

OBJECT	LAUNCHED	DECAYED	LAST ORBIT (KM)
Delta 2 first stage, NAVSTAR 18 GPS mission (20453)	**Jan 24, 1990**	**Oct 10**	**154 × 161**
Pegasus upper stage, CYGNSS (41892)	Dec 15, 2016	Oct 10	180 × 190
Electron upper stage, "Owl Around The World" (100562)	Sep 2, 2026	Oct 10	145 × 196

Catalog numbers link to CelesTrak records; orbits are the last perigee and apogee in the catalog.



A SpaceX cargo Dragon backs away from the International Space Station after undocking from the Harmony module's forward port on June 16, 2026, above a cloudy North Pacific.

PHOTO: NASA · PUBLIC DOMAIN

Station traffic

CRS-35 is due to dock on its own at the space-facing port of Harmony, vacated by Crew-12's Dragon last week, on Thursday, October 15, and to stay until mid-November before splashing down off California (NASA). The next Russian cargo ship, Progress MS-36, is listed for 12:03 a.m. Eastern Standard Time (EST) on November 24 from Baikonur (Launch Library 2).

V · INDUSTRY & REGULATION

Starship comments due Wednesday; launch rules open to November 4

The proposals target Part 450, the licensing rule for U.S. commercial launches and reentries in effect since 2021. Separate environmental reviews could allow many more Starship and New Glenn launches from Florida.

Part 450 proposals

Five notices of proposed rulemaking appeared in the Federal Register on October 5, with comments due November 4. They would simplify how operators show compliance with flight safety analysis (docket FAA-2026-9936); add an exception to lightning-hazard rules (FAA-2026-9940); delete the word "hypothetical" from far-field blast analysis, so it covers existing windows only (FAA-2026-9941); simplify physical containment

requirements (FAA-2026-9934); and allow fully digital license applications (FAA-2026-9935).

Transportation Secretary Sean Duffy announced the package, the first output of the Department of Transportation's Space Policy Administration, Coordination, and Execution (SPACE) Task Force, Payload reported. The changes are meant to help reach the White House goal of 1,000 commercial launches and reentries a year by 2030. Acting Deputy FAA Administrator Liam McKenna called them "just the beginning."

Environmental reviews in Florida

Starship at SLC-37. The FAA is preparing a Supplemental Environmental Impact Statement (SEIS) on SpaceX's plan for up to 76 Starship launches and 152 landings a year at Cape Canaveral. Scoping comments are due this Wednesday, October 14 (notice, docket FAA-2026-10792).

New Glenn at SLC-36. A draft Environmental Assessment (EA) covers Blue Origin's request to raise New Glenn's cap from 12 to 50 launches a year, with no new construction. A virtual public meeting is set for October 22 from 6 to 8 p.m. EDT, and comments close October 30 (notice, docket FAA-2026-9841).

Amazon Leo nears service

Amazon has built its 1,000th Leo broadband satellite and has 396 in orbit, and it plans initial service in 2026, the company told Ars Technica last week. Its next launches are the Vulcan and Ariane 64 flights in the schedule above.

QUICK HIT

Terran R. The FAA has adopted a Finding of No Significant Impact and Record of Decision for Relativity Space's Terran R program at Cape Canaveral, under docket FAA-2026-12046 (notice).

Europe

Themis. ArianeGroup and SSC Space completed a second wet dress rehearsal of the Themis reusable-rocket demonstrator at Esrange, Sweden, on September 29. The vehicle was filled with liquid nitrogen instead of propellant. ESA announced the test on October 7 (European Spaceflight). The European Union has extended its €42.8 million SALTO (reuSable strAtegic space Launcher Technologies & Operations) project, which funds Themis's first two hop tests, to December 31, 2026 (European Spaceflight).

HyImpulse. The German startup received a launch operator license from the UK Civil Aviation Authority (CAA) on October 1. The license covers its SR75 suborbital rocket from SaxaVord, Scotland, but the flight has slipped to spring 2027. The company cited the remaining licensing work and autumn weather (European Spaceflight).

India

Agnikul Cosmos fired its single-piece, 3D-printed Agnite semi-cryogenic engine. The test included thrust sweeps and a run "well beyond the max. burn time of some of our ascent phase engines," as the company prepares for its Agnibaan Mission-02 (India Today).

QUICK HIT

Blue Origin in Texas. Blue Origin announced Constellation Park in Hutto, Texas, a 1.3 million-square-foot site with 2,000 jobs, for its TeraWave and Quartz programs.

VI · BY THE NUMBERS

239 orbital launch attempts so far in 2026

SpaceX accounts for almost half. China needs at least 30 more launches to pass 100 in a year for the first time.

239	232	116	71	30
orbital launch attempts worldwide, Jan 1–Oct 11	reached orbit	by SpaceX, all successful	by China, four failures	more Chinese launches needed this year to pass 100

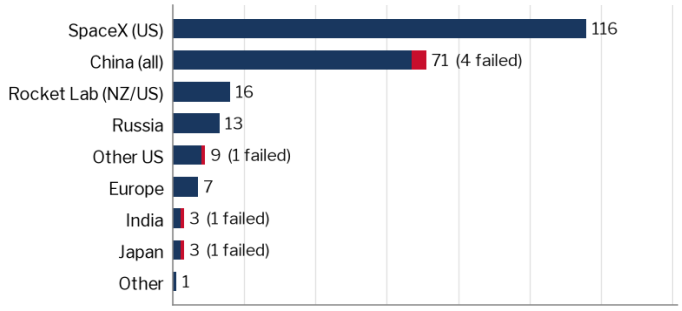
The count

Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) records 239 orbital launch attempts through October 10, 232 of them successful, including Friday's Long March 12 and Saturday's Falcon 9. SpaceX has flown 116, all successful, followed by China with 71, Rocket Lab's Electron from New Zealand with 15 (plus one from the U.S.) and Russia with 13.

Busiest launch sites

Vandenberg Space Force Base in California is the year's busiest launch site, ahead of Florida's Cape Canaveral and Kennedy Space Center combined. March and April were the busiest months, with 30 orbital launch attempts each (GCAT, through October 10).

SITE	LAUNCHES, 2026
Vandenberg Space Force Base, California	**64**
Cape Canaveral and Kennedy Space Center, Florida	56
Jiuquan, China	26
Mahia, New Zealand (Rocket Lab)	15
Hainan commercial launch site, Wenchang, China	14
Taiyuan, China	12
Plesetsk, Russia	8



Orbital launch attempts in 2026 by provider or country. Navy: success. Red: failure or partial failure.

CHART: THE LAUNCH REPORT · GCAT (JONATHAN MCDOWELL) THROUGH OCT 10

The failures

According to GCAT, seven launches have failed this year: India's Polar Satellite Launch Vehicle (PSLV) on January 12; China's Long March 3B on January 16, Ceres-2 on January 17, Tianlong-3 on April 3 and Long March 7A on August 10; Japan's Kairos on March 5; and Blue Origin's New Glenn on April 19, recorded as a partial failure. A New Glenn that exploded on its pad on May 29 is not counted as a launch attempt.

China's fourth-quarter push

China reached 71 orbital launch attempts with Friday's Long March 12. To pass 100 in a year for the first time, it needs at least 30 more by December 31; it flew 18, 26 and 26 in the first three quarters, SpaceNews reported. Possible first flights include Orienspace's Gravity-2, iSpace's Hyperbola-3, which would attempt first-stage recovery, and Zenk Space's Zhihang-1.

KEY DATES AHEAD (EDT)

Mon Oct 12	Falcon 9 Starlink 15-25, 8:11 p.m.
Tue Oct 13	CRS-35 launch, 6:33 a.m.
Wed Oct 14	Starship SLC-37 scoping comments due
Thu Oct 15	CRS-35 docks; USSF-481, 7:44 p.m.; NVS-03 window opens
Mon Oct 19	H3 / MMX, 3:41 p.m.
Thu Oct 22	New Glenn cadence meeting, 6–8 p.m.
NET Oct 29	Vulcan returns with Amazon Leo

Takeaways

1. Falcon 9 launched 21 SDA satellites at 3:39 a.m. EDT Saturday after a week of delays.
2. A Long March 12 put China's 27th SatNet group in orbit on Friday, China's 71st launch of 2026.
3. All three rocket stages predicted for Saturday, including a 1990 Delta 2 stage, reentered.
4. CRS-35 launches at 6:33 a.m. EDT Tuesday; H3 carries MMX toward Mars on October 19.
5. Musk said Starship Flight 15, with a first ship-catch attempt, is four to six weeks away.

SOURCES

Schedules: Launch Library 2 (The Space Devs), Spaceflight Now, NASASpaceflight. Agencies: NASA, JAXA, FAA, Federal Register. Tracking: CelesTrak, GCAT (Jonathan McDowell). Reporting: Xinhua, TSN Media, SpaceNews, Ars Technica, Payload, European Spaceflight, Times of India, India Today. X posts: @blueorigin. Photos: NASA (public domain). Maps: Natural Earth (public domain).

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